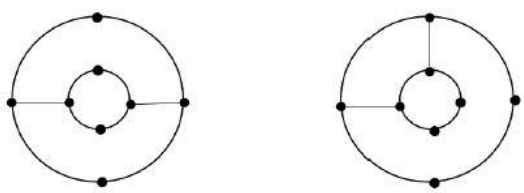
MAT 206 GRAPH THEORY

QUESTION BANK

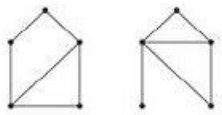
UNIT WISE QUESTION BANK

1. Prove that in a connected graph every $u - v$ walk contains a $u - v$ path.
2. Prepare a short note on walk, trial, path, and cycle of a connected graph.
3. Draw all simple graph of one, two, three, and for vertices.
4. Draw the graph of the following chemical compounds CH_4, C_2H_6, C_6H_6

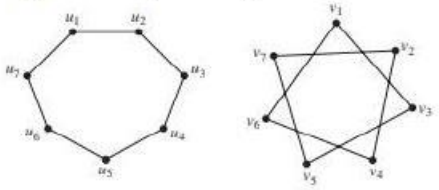
5. Show that the maximum number of edges in a simple graph with 'n' vertices is $n(n-1)/2$.
6. Prove that the maximum number of edges in a simple graph with 'n' vertices is $(n-1)$.
7. Draw a connected graph that becomes disconnected when any edge is removed from it.
8. Prove that a simple graph with "n" vertices must be connected if it has more than $(n-1)(n-2)/2$ edges.
9. Are the two graphs in the figure isomorphic.



10. Prove that any two simple connected graphs with 'n' vertices all of degree 2 are isomorphic.
11. What is the maximum number of edges in a simple graph with 'n' vertices?
12. There are 25 telephones in Metropolis. Is it possible to connect them with wires so that each telephone is connected with exactly 7 others? Why?
13. Define complete graph and complete bipartite graph. Draw a graph which is a complete graph as well as a bipartite graph.
14. Explain walk, path, and circuit with the help of examples.
15. Define isolated vertex, pendent vertex, even vertex and odd vertex. Draw a graph that contains all the above.
16. What is the number of vertices in an undirected connected graph with 27 edges,6 vertices of degree 2,3 vertices of degree 4, and the remaining of degree3?
17. Determine whether the graphs are isomorphic. If Yes or No justify



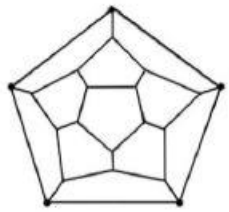
18. Determine whether the given pairs of graphs is isomorphic. Exhibit an isomorphism or provide a rigorous argument that none exists.



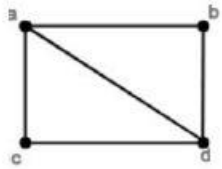
19. Explain seating problem in graph theory using graphical representation.
20. Can five houses be connected to two utilities without connections crossing? Justify your answer by drawing a graph.

UNIT WISE QUESTION BANK

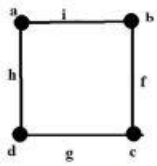
1. Is Hamiltonian circuit always implying Hamiltonian path? Is the converse true? Justify your answer.
2. Is the converse of Dirac's theorem true? Justify your answer.
3. Prove that if a connected graph G is decomposed into two sub graphs g_1 and g_2 there must be at least one vertex common between g_1 and g_2.
4. Is Peterson graph Hamiltonian. Justify your answer.
5. Does the following graph have a Hamiltonian Circuit?

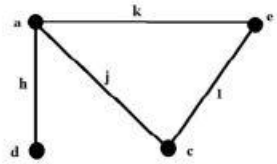


6. Check whether the graph is Eulerian graph. Justify your answer.



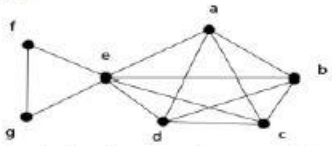
7. Find the union, intersection and ring sum of the following graphs.



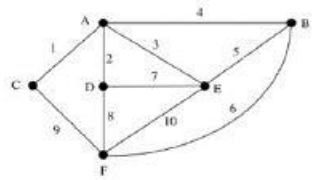


8. State travelling salesman problem. How it is related to Hamiltonian circuits?
9. For which values of m, n is the complete graph k_{mn} an Euler graph ? Justify your answer.
10. State Dirac's theorem and check the applicability of the following graph.

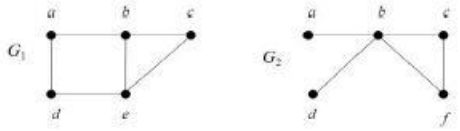
11. Define an Euler graph. Check whether the given graph is Euler graph and if yes, give the Euler walk. Justify your answer.



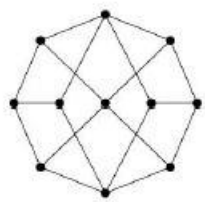
12. Find the smallest integer n such that the complete graph has at least 500 edges.
13. Draw a connected graph that becomes disconnected when any edge is removed is removed from it.
14. Define a Hamilton Cycle. Illustrate. Describe the Travelling Sales man Problem.
15. Fuse the vertices A and B of the following graph



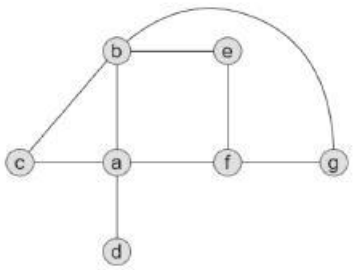
16. Find the union, intersection, and ring sum of the following two graphs



17. For what value of 'n' do the complete graph K_n have an Euler circuit? Give the reason.
18. Consider the following graph, find a Hamiltonian path by labelling the vertices of the given graph, can your path be extended to a Hamiltonian circuit?

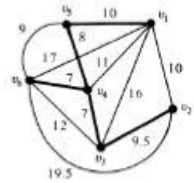


19. (a) Using figurative representation and explain the difference between complete symmetric digraph and complete asymmetric digraph. (b) When are two digraphs considered to be the same or isomorphic, explain with example.
20. Use Fleury's Algorithm to find Euler path or Euler Circuit in the following graph

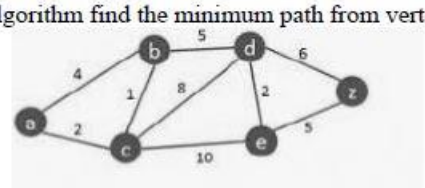


UNIT WISE QUESTION BANK

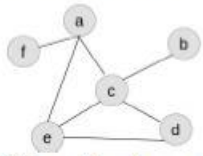
1. Draw all non-isomorphic trees with seven vertices.
2. What is the nullity of a complete graph with n vertices?
3. Show that Hamiltonian path is a spanning tree.
4. Using Kruskal's algorithm find the minimum spanning tree



5. Using Dijkstra's algorithm find the minimum path from vertex a to z.

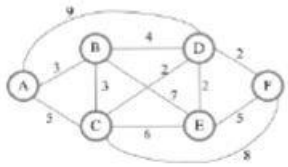


6. Find the Eccentricity, Diameter, Radius and Centre of the graph shown below:

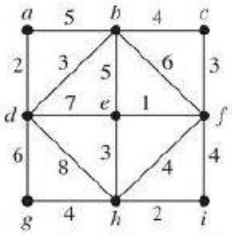


7. Prove that every connected graph has at least one spanning tree.

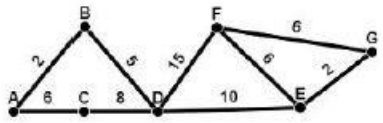
8. Explain Dijkstras Algorithm and find the shortest path from A to F.



9. Prove that a tree with n vertices has $n-1$ edges.
10. Prove that a pendant edge in a connected graph G is contained in every spanning tree of G
11. Cite three different situations that can be represented by trees. Explain.
12. Prove that any circuit in a graph G must have atleast one edge in common with a chord set.
13. How many isomers does Pentane C_5H_{12} have?
14. Prove that two colours are necessary and sufficient to paint all n vertices $(n \geq 2)$ of a tree, such that no edge in the tree has both of its end vertices of the same colour
15. Find the maximum and minimum level of a rooted binary tree with 75 vertices
16. Use Prim's algorithm to find a minimum spanning tree of the weighed graph given below

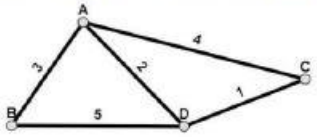


17. Describe Dijkstra's algorithm and find the shortest path from a to z in the following graph.



18. Describe Kruskal's algorithm and apply it to find the minimal spanning tree for the following graph.

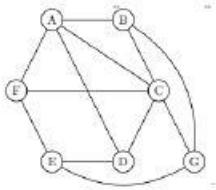
19. Describe Floyd Warshall algorithm and use it to find shortest distance between all pairs of vertices in the following graph



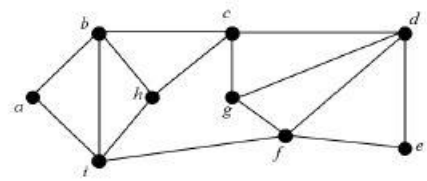
20. Define branches and chords of a graph. Find the number of chords in a graph with 17 vertices and 24 edges.

UNIT WISE QUESTION BANK

1. Define edge connectivity, vertex connectivity and separable graphs. Give an example for each.
2. Prove the statement using an example: Every cut set in a connected graph G must also contain at least one branch of every spanning tree of G.
3. State Kuratowski's theorem and use it to show that the graph G below is not planar. Draw G on the projective plane without edges crossing. Your drawing should use the labelling of the vertices given.

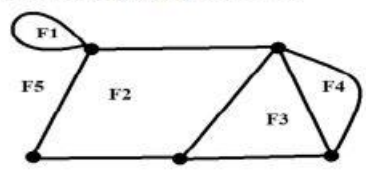


4. Let G be a connected graph and e an edge of G. Show that *e* is a cut-edge if and only if *e* belongs to every spanning tree.
5. If G is a 5-regular simple graph and |V| = 10, prove that G is non-planar.
6. State and prove Euler's Theorem relating the number of faces, edges and vertices for a planar graph.
7. From the following graph form a spanning tree and give all the fundamental cut set and fundamental circuits.

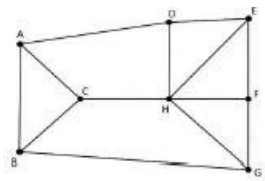


8. Define fundamental circuit and fundamental cut-sets.

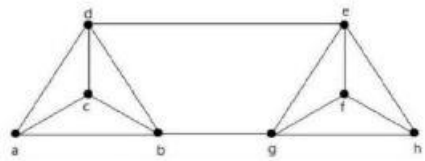
9. Show that a set of fundamental circuits in a planar graph G corresponds to a set of fundamental cut sets in its dual G^*.
10. Define planar graphs. Is K_4, the complete graph with 4 vertices, a planar graph?
11. Define cut-set. Prove that every circuit in G has an even number of edges in common with any cut-set.
12. Construct the geometric dual of the graph given below



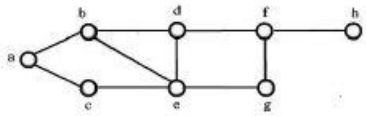
13. Prove that a connected planar graph with n vertices and e edges has $e - n + 2$ regions.
14. Let G be a connected graph and e an edge of G. Show that e is a cut-edge if and only if e belongs to every spanning tree.
15. Prove that in any simple, connected planar graph with f region, n vertices, and e edges $(e > 2)$, the following inequalities must hold $e \geq \frac{3}{2}f$, $e \leq 3n - 6$.
16. Define fundamental cut set of a graph. Find all fundamental cut set of the graph below with respect to a spanning tree.



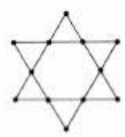
17. Calculate the vertex connectivity edge connectivity and minimum vertex degree of the following graph.



18. How many vertices does a connected planar graph G have if it has 7 edges and can be drawn without any edge crossings with 4 faces?
19. With respect to a spanning tree for the following graph find all fundamental circuits.

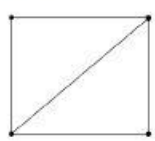


20. Draw a geometric dual of the graph

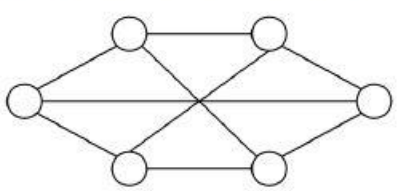


UNIT WISE QUESTION BANK

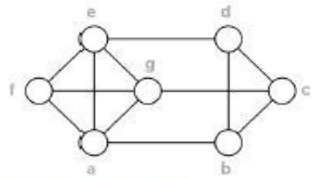
1. Using induction on 'n', prove that an 'n-vertex' is a tree if and only if its chromatic polynomial is $P_n(\lambda) = \lambda(\lambda - 1)^{n-1}$.
2. Sketch two different (non-isomorphic) graphs that have the same chromatic polynomial.
3. Find the chromatic polynomial of the graph



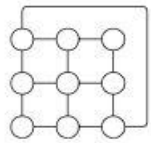
4. Find chromatic number of the following graph



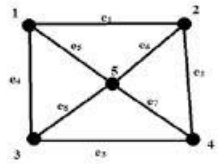
5. Find chromatic number of the following graph by Greedy Algorithm



6. Find chromatic number of the following graph

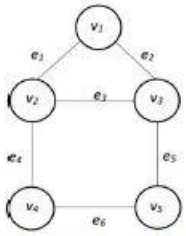


7. Construct the adjacency matrix and incidence matrix of the following graph

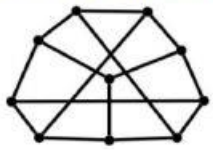


8. Define chromatic number. What is the chromatic number of a tree with two or more vertices?
9. A graph with at least one edge is 2-chromatic if and only if it has no circuits of odd length.
10. Prove that a graph with at least one edge is 2-chromatic if and only if it has no circuits of odd length.
11. Prove that a graph of n vertices is a complete graph iff its chromatic polynomial is $P_n(\lambda) = \lambda(\lambda - 1) \dots (\lambda - n + 1)$.
12. Explain four colour problem using the concept of chromatic number.

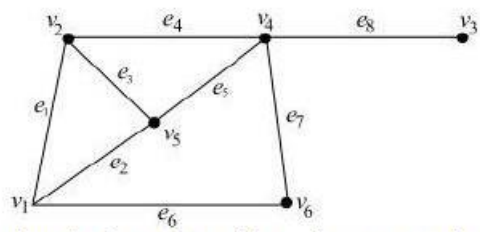
13. Construct incidence matrix for the following graph



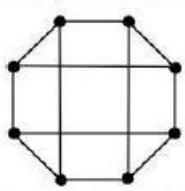
14. What will be the chromatic number for the following graph?



15. Prove that an n −vertex graph is a tree iff its chromatic polynomial $P_n(\lambda) = \lambda(\lambda-1)^{(n-1)}$
16. Construct the adjacency matrix X for the following graph and write properties of adjacency matrix



17. Colour the faces of K_4 with 4 colours. Can only 3 colours be used?
18. What will be the chromatic number for the following graph?



19. Prove that a graph with at least one edge is 2-chromatic if and only if it has no circuits of odd length.
20. Let B and A be respectively the circuit matrix and the incidence matrix of a self-loop free graph, whose columns are arranged using the same order of edges. Then show that $AB^T = BA^T \equiv 0(mod2)$.